

Curriculum Vitae

Personal details

Name

Date of birth

Email

Mobile

Address

LUONG CAO THANG

March 26, 1989

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21 L Street, Nhi Dong 2, Di An Ward, Di An City, Binh Duong Province, Vietnam



Education

University

Period

Degree

Major

Cumulative GPA

Thesis

Tampere University of Technology

2012-2015

Master's degree in Electrical Engineering

Wireless Communication Circuits and Systems

4.04/5

Development of I²C Adapter: Implementing the digital logic on I²C protocol to communicate between a capacitive LCD multimedia color touch panel and Dual-core ARM Cortex-A9 in an FPGA

University

Period

Degree

Major

Cumulative GPA

Achievements

Thesis

Vietnam National University Ho Chi Minh City – International University

2007 – 2012

Bachelor's degree in Electrical Engineering

Electronic

80.1/100

Graduated with high honors from the Faculty of Electrical Engineering in 2012
Digital Logic Implementation of DNA Comparison Algorithm: Implementing the digital logic to perform a DNA comparison based on the Smith-Waterman algorithm in an FPGA

Working Experience

Title

Company

Period

Responsibilities

Senior Physical Design Engineer

MARVELL

January 2023 – March 2025

Projects:

- Pleiades (TSMC 16nm): Responsible for two blocks with good QoR delivery

- Alpine (TSMC 12nm): PCIe Gen5 based SSD-Controller

Block NVME, 5 million instances

- Applying cluster to the endpoints of the latch cell to resolve timing violation related to the latch cell

- Applying scripting to create the obstruction on the cell to resolve short issues related to the multibit flop

- Lascaux (TSMC 5nm): block rel_wrap, 2 million instances

- The block is highly sensitive to congestion. Experimenting with various floorplans and partial blockages to resolve the congestion.

- Placing boundary clock divider cells close to clock port and shortening the timing to the clock divider cells to resolve the timing violations.

Title
Company
Period
Responsibilities

Physical Design Engineer
SEMIFIVE
June 2021 – December 2022
- Training fresher in basic Place & Route knowledge and block implementation
Projects:
- RHEA (Samsung 14nm): Block Mainbus_Subsystem, 1.5 million instances
• The narrow block shape causes routing congestion and noise
• Routing at the higher layers, applying NDR and inserting buffers to fix the issue
- ATOM (Samsung 5nm): Block PERI_Subsystem, 200 thousand instances
• The complex clock structure causes significant skew
• Create relative placement groups & apply clock skew groups

Title
University
Period
Responsibilities

Researcher in Analog Mixed Signal Design
Technical University of Berlin - Germany
April 2017 – February 2021
- Design Pipelined Analog to Digital Converter in TSMC 65 nm technology
• Target: 14-bits 1 GS/s Time-Interleaved Pipelined ADC
• Full scale range: 1.8 V_{p-p,diff}
• Nyquist SFDR: 75.78 dBc
• Power (ADC, reference generator, bias and digital): 120 mW
- Layout basic blocks in TSMC 65nm and GF 22nm technology
- Teaching assistant for Analog Integrated Circuits subjects

Title
University
Period
Responsibilities

Lecturer
Posts and Telecommunications Institute of Technology (PTIT)
July 2016 – February 2017
- Lecturing on subjects in electrical engineering
- Organize activities and seminars in electrical science

Title
Company
Period
Responsibilities

Product Development Engineer
Intel Products Vietnam
June 2015 – May 2016
- Responsible for ensuring the testability and manufacturability of integrated circuits from the component feasibility stage through production ramp

Title
Company
Period
Responsibilities

Internee in Embedded System
Vietnam National University Ho Chi Minh City - Integrated Circuit Design Research and Education Center
May 2011 – November 2011
- FPGA Design

Technical

Languages: VHDL, Verilog, C/C++, Perl, Tcl, Matlab
Tools: Familiar to use

- Synopsys: **Fusion Compiler, ICC2, PrimeTime**, Design DA
- Cadence: **Innovus**, Virtuoso
- Mentor: **Calibre** (DRC/LVS)
- Altera: Quartus II, ModelSim, Altera Monitor Program

Language

Vietnamese (native), English (Advanced – IELTS), Finnish (SUOMI 1), German (B1)